

IBPS SO HR — Quantitative Aptitude Study Guide

Purpose

This guide is designed for an **MA HRM student preparing for IBPS SO HR/Personnel Officer**.

The goal is not to become an advanced mathematician. The goal is to:

- understand the basic idea behind each Quant topic
 - know the important formulas
 - solve common banking-exam question types
 - become faster and more accurate
 - recognise shortcuts and common traps
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How to Use This Guide

For every topic, follow this cycle:

Learn → Understand → Solve Basic Questions → Solve Exam Questions → Time Yourself → Record Mistakes → Revise

Do not try to memorise every shortcut before understanding the basic method.

A useful rule:



Accuracy first. Speed second.

Target roughly **80–85% accuracy** on a topic before moving to the next one.

Quant Roadmap

Mathematical Basics
↓
Simplification + Approximation
↓
Percentage
↓
Ratio & Proportion
↓
Average
↓
Profit, Loss & Discount
↓
Simple Interest + Compound Interest
↓
Partnership
↓
Time & Work
↓
Pipes & Cisterns
↓
Time, Speed & Distance
↓
Trains
↓
Boats & Streams
↓
Mixture & Allegation
↓

Ages



Number System



HCF + LCM + Remainders



Quadratic Equations



Number Series



Data Interpretation



Data Sufficiency



Probability



Permutation & Combination



PYQs → Sectional Tests → Full Mocks

PART 0 — Mathematical Foundation

0.1 Number Basics

What is it?

Before solving Quant, you should know the basic types of numbers.

- **Natural numbers:** 1, 2, 3, ...

- **Whole numbers:** 0, 1, 2, 3, ...
- **Integers:** ..., -2, -1, 0, 1, 2, ...
- **Even numbers:** divisible by 2
- **Odd numbers:** not divisible by 2
- **Prime number:** exactly two factors — 1 and itself
- **Composite number:** more than two factors

Tip

Remember:

- 1 is **neither prime nor composite**.
 - 2 is the **only even prime number**.
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0.2 Fractions and Decimals

A fraction tells us a part of a whole.

Example:

$$\frac{3}{4} = 0.75 = 75\%$$

Useful conversions:

Fraction	Percentage
1/2	50%
1/3	33.33%
1/4	25%
1/5	20%
1/8	12.5%
1/10	10%

1/20	5%
1/25	4%

Trick

Memorise common fraction-to-percentage conversions. They make Percentage and DI much faster.

0.3 BODMAS / Simplification

Order:

Brackets

Orders / powers

Division

Multiplication

Addition

Subtraction

Example:

$$10 + 5 \times 2$$

First multiplication:

$$10 + 10 = 20$$

Not 30.

Tip

Division and multiplication are handled from **left to right**.

The same applies to addition and subtraction.

0.4 Algebra Basics

You will often see an unknown represented by x .

Example:

$$x + 5 = 12$$

Therefore:

$$x = 7$$

Important identities:

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

Trick

Do not memorise identities without understanding them. Use them repeatedly in simplification and quadratic questions.

PART 1 — Simplification

What is Simplification?

You are given a mathematical expression and asked to find its value.

Example:

$$25 + 5 \times 4$$

Answer:

$$25 + 20 = 45$$

What to practise

- Fractions
- Decimals
- Percentages
- Squares
- Square roots
- Basic algebra
- BODMAS

Shortcut

Know:

- squares from 1 to 30 at minimum
- common fraction values
- multiplication tables up to 20 or 30

Common trap

Do not calculate left-to-right blindly. Follow BODMAS.

PART 2 — Approximation

What is Approximation?

Instead of calculating an exact answer, you use a nearby easy number.

Example:

$$49.8 \times 20.1$$

Can often be treated approximately as:

$$50 \times 20 = 1000$$

This is useful when answer options are far apart.

Tricks

If:

- $49.8 \approx 50$
- $19.9 \approx 20$
- $1001 \approx 1000$

use the simpler value when the question clearly allows approximation.

Exam Tip

Approximation is mainly a **speed tool**.

Do not approximate when answer options are very close.

PART 3 — Percentage

What is Percentage?

Percentage simply means "**out of 100.**"

$$25\% = 25/100 = 1/4$$

Basic Formula

$$\text{Percentage} = (\text{Part} / \text{Whole}) \times 100$$

Example:

A student scores 60 out of 80.

$$60/80 \times 100 = 75\%$$

Percentage Increase

$$\text{Increase \%} = (\text{Increase} / \text{Original Value}) \times 100$$

Example:

Salary rises from ₹20,000 to ₹24,000.

$$\text{Increase} = ₹4,000$$

$$4000/20000 \times 100 = 20\%$$

Percentage Decrease

$$\text{Decrease \%} = (\text{Decrease} / \text{Original Value}) \times 100$$

Successive Percentage Change

If a value increases by $a\%$ and then by $b\%$:

$$\text{Net change} = a + b + ab/100$$

Example:

20% increase followed by 10% increase:

$$20 + 10 + (20 \times 10)/100 = 32\%$$

Increase followed by decrease

20% increase followed by 10% decrease:

$$20 - 10 - (20 \times 10)/100 = 8\% \text{ increase}$$

Important Trap

A 20% increase followed by a 20% decrease is **not zero**.

Using the formula:

$$20 - 20 - 400/100 = -4\%$$

So there is a **4% decrease**.

Percentage ↔ Fraction Trick

Memorise:

12.5% = $\frac{1}{8}$
16.67% $\approx \frac{1}{6}$
20% = $\frac{1}{5}$
25% = $\frac{1}{4}$
33.33% $\approx \frac{1}{3}$
37.5% = $\frac{3}{8}$
50% = $\frac{1}{2}$
62.5% = $\frac{5}{8}$
75% = $\frac{3}{4}$
80% = $\frac{4}{5}$

PART 4 — Ratio and Proportion

What is Ratio?

Ratio compares two quantities.

If there are 2 boys and 3 girls:

Boys : Girls = 2 : 3

It does not mean there are only 2 boys and 3 girls. It tells us the **relationship**.

Simplifying Ratio

20 : 30

Divide both by 10:

2 : 3

Proportion

Two ratios are equal.

2 : 3 = 4 : 6

Direct Proportion

When one quantity increases, the other also increases.

Example:

More workers → more work completed in the same time.

Inverse Proportion

When one quantity increases, the other decreases.

Example:

More workers → fewer days required for the same work.

Trick

Ask:

"If A increases, should B increase or decrease?"

That usually tells you whether the relationship is direct or inverse.

PART 5 — Average

What is Average?

Average is the **equal share** of all values.

$\text{Average} = \text{Total} / \text{Number of items}$

Therefore:

$\text{Total} = \text{Average} \times \text{Number of items}$

This second formula is extremely useful.

Example

Five employees have salaries averaging ₹30,000.

Total salary:

$30,000 \times 5 = ₹1,50,000$

Replacement Trick

If one value is removed and another added:

$$\text{Change in total} = \text{New value} - \text{Old value}$$

Then:

$$\text{Change in average} = \text{Change in total} / \text{Number of items}$$

Common Trap

Use the **original number of items** if one value is replaced by another.

PART 6 — Profit, Loss and Discount

Basic Terms

- **CP** = Cost Price
- **SP** = Selling Price
- **MP** = Marked Price

Profit

If $SP > CP$:

$$\text{Profit} = SP - CP$$

$$\text{Profit \%} = \text{Profit} / CP \times 100$$

Loss

If $CP > SP$:

$$\text{Loss} = CP - SP$$

$$\text{Loss \%} = \text{Loss} / CP \times 100$$

Discount

$$\text{Discount} = MP - SP$$

$$\text{Discount \%} = \text{Discount} / MP \times 100$$

Very Important

Profit percentage is calculated on **CP**.

Discount percentage is calculated on **MP**.

That distinction causes many mistakes.

Successive Discount

If discounts are $a\%$ and $b\%$:

$$\text{Net discount} = a + b - \frac{ab}{100}$$

Example:

20% and 10%:

$$20 + 10 - 2 = 28\%$$

Not 30%.

PART 7 — Simple Interest

Formula

$$SI = P \times R \times T / 100$$

Where:

- P = Principal
- R = Rate
- T = Time

$$\text{Amount} = \text{Principal} + SI$$

Example

₹10,000 at 10% per year for 2 years:

$$SI = 10000 \times 10 \times 2 / 100$$

$$= ₹2,000$$

Amount:

$$₹12,000$$

Trick

For simple interest, interest is always calculated on the **original principal**.

PART 8 — Compound Interest

What is Compound Interest?

Interest is added to the amount, and the next period earns interest on the larger amount.

Formula:

$$A = P (1 + R/100)^T$$

$$CI = A - P$$

Quick Trick

For 2 years:

$$CI - SI = P \times (R/100)^2$$

Example:

$$P = ₹10,000, R = 10\%$$

Difference:

$$10,000 \times (10/100)^2$$

$$= ₹100$$

Important

Know the difference between:

- annual compounding
- half-yearly compounding
- quarterly compounding

For half-yearly:

- rate becomes $R/2$
 - time becomes $2T$
-

PART 9 — Partnership

Basic Idea

Profit is divided according to:

$\text{Investment} \times \text{Time}$

If A invests ₹10,000 for 12 months and B invests ₹20,000 for 6 months:

A's contribution:

$$10,000 \times 12 = 1,20,000$$

B's:

$$20,000 \times 6 = 1,20,000$$

So profit ratio:

1 : 1

Trick

Do not compare investment alone. Compare:

Capital × Time

PART 10 — Time and Work

Basic Idea

If a person completes a job in 10 days:

$$1 \text{ day work} = 1/10$$

If another person takes 20 days:

$$1 \text{ day work} = 1/20$$

Together:

$$1/10 + 1/20 = 3/20$$

Therefore time:

$$20/3 \text{ days}$$

Golden Formula

If:

- A takes x days
- B takes y days

Together:

$$1/x + 1/y$$

Efficiency Trick

Efficiency and time are inversely related.

If A is twice as efficient as B:

A takes **half the time**.

PART 11 — Pipes and Cisterns

This is basically **Time and Work with water**.

- Filling pipe = positive work
- Emptying pipe = negative work

Example:

Pipe A fills a tank in 10 hours.

Rate:

$$1/10$$

Pipe B empties it in 20 hours.

Rate:

$-1/20$

Together:

$$1/10 - 1/20 = 1/20$$

Tank fills in 20 hours.

Trick

Always decide whether the pipe is:

adding water (+) or removing water (-).

PART 12 — Time, Speed and Distance

Golden Triangle

$$\frac{\text{Distance}}{\text{Speed} \times \text{Time}}$$

Therefore:

$$\text{Speed} = \text{Distance} / \text{Time}$$

$$\text{Time} = \text{Distance} / \text{Speed}$$

$$\text{Distance} = \text{Speed} \times \text{Time}$$

Unit Conversion

Very important:

$$1 \text{ km/h} = 5/18 \text{ m/s}$$

$$1 \text{ m/s} = 18/5 \text{ km/h}$$

Trick

For km/h $\rightarrow$ m/s:

multiply by 5/18

For m/s $\rightarrow$ km/h:

multiply by 18/5

Average Speed

Do NOT simply average speeds unless distances are equal.

For equal distances:

$$\text{Average speed} = 2xy / (x+y)$$

where x and y are the two speeds.

PART 13 — Trains

Important relationships:

$$\text{Time} = \text{Distance} / \text{Speed}$$

When a train crosses:

A pole/person

Distance = **length of train**

A platform

Distance = **train length + platform length**

Two trains

Distance = **sum of train lengths**

Trick

Always ask:

"What total distance is the moving object covering?"

That prevents many mistakes.

PART 14 — Boats and Streams

Let:

- Boat speed in still water = B
- Stream speed = S

Then:

$$\text{Downstream speed} = B + S$$

$$\text{Upstream speed} = B - S$$

Therefore:

$$B = (\text{Downstream} + \text{Upstream}) / 2$$

$$S = (\text{Downstream} - \text{Upstream}) / 2$$

Memory Trick

Downstream → Add

Upstream → Subtract

PART 15 — Mixture and Allegation

Basic Idea

You mix two quantities with different prices or concentrations.

Example:

₹20/kg rice + ₹30/kg rice.

If mixture price is ₹25/kg, the quantities will normally be in equal ratio.

Alligation

Think of it as a **cross-difference method**.

Cheap 20	Mean 25	Expensive 30
	$30 - 25 = 5$	
	$25 - 20 = 5$	

Ratio:

$$5 : 5 = 1 : 1$$

Trick

For two quantities:

Quantity of cheaper = expensive price - mean price

Quantity of expensive = mean price - cheaper price

PART 16 — Problems on Ages

Basic Idea

Age changes by the same number of years for everyone.

If A is 5 years older than B:

After 10 years, A is still **5 years older**.

Important

Differences stay constant.

Ratios change.

Example:

$$A = 20, B = 10$$

$$\text{Ratio} = 2 : 1$$

After 10 years:

$$30 : 20 = 3 : 2$$

Trick

When the problem gives a ratio, convert the ages into:

kx and mx

Then form an equation.

PART 17 — Number System

Important Concepts

Study:

- Divisibility
- Factors

- Multiples
 - Prime numbers
 - Composite numbers
 - Remainders
 - Unit digit
 - Last digits
-

Divisibility Rules

By 2

Last digit is even.

By 3

Sum of digits is divisible by 3.

By 4

Last two digits are divisible by 4.

By 5

Last digit is 0 or 5.

By 6

Divisible by both 2 and 3.

By 8

Last three digits divisible by 8.

By 9

Sum of digits divisible by 9.

By 10

Last digit is 0.

By 11

Difference between sums of alternating digits is divisible by 11.

PART 18 — HCF and LCM

HCF

Highest Common Factor.

It is the largest number that divides the given numbers exactly.

LCM

Least Common Multiple.

It is the smallest number divisible by all given numbers.

Important Relationship

For two positive integers:

$$\text{HCF} \times \text{LCM} = \text{Product of the two numbers}$$

Trick

For fraction-related questions:

- HCF of fractions often uses **HCF of numerators / LCM of denominators**
- LCM of fractions often uses **LCM of numerators / HCF of denominators**

Use carefully and understand the structure before applying.

PART 19 — Remainders

Basic Idea

When a number is divided:

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

The remainder is always smaller than the divisor.

Unit Digit / Cyclicity

Powers often repeat their last digit.

Example for 2:

$$2^1 \rightarrow 2$$

$$2^2 \rightarrow 4$$

$$2^3 \rightarrow 8$$

$$2^4 \rightarrow 6$$

$$2^5 \rightarrow 2$$

Cycle length = 4.

Trick

For large powers:

1. Find the repeating cycle.
2. Find the position of the required power inside the cycle.
3. Use that last digit.

PART 20 — Quadratic Equations

Basic Idea

An equation like:

$$x^2 - 5x + 6 = 0$$

can be factorised:

$$(x-2)(x-3) = 0$$

So:

$$x = 2 \text{ or } 3$$

IBPS Use

Questions often ask you to solve two equations and compare their roots.

Trick

Learn quick factorisation.

For:

$$x^2 - 7x + 12$$

Find two numbers whose:

- product = 12
- sum = 7

Answer:

$$3 \text{ and } 4$$

So:

$$x = 3, 4$$

PART 21 — Number Series

What is it?

You are given a sequence and asked to find:

- missing number
- wrong number
- next number

Example:

2, 4, 8, 16, ?

Pattern:

$\times 2$

Answer:

32

Patterns to Check

Look in this order:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Squares
6. Cubes
7. Alternating pattern
8. Increasing differences
9. Mixed operations

Trick

Don't assume the pattern is multiplication immediately.

Check the **differences between consecutive terms** first.

PART 22 — Data Interpretation (DI)

What is DI?

You receive data in a:

- table
- graph
- chart
- paragraph

and answer questions from it.

DI is not a completely new topic.

It combines:

Percentage + Ratio + Average + Arithmetic + Calculation speed

22.1 Table DI

Data appears in rows and columns.

Practice:

- totals
- differences
- percentages
- ratios
- averages

Trick

Do not calculate the entire table.

Read the question first and calculate only what you need.

22.2 Bar Graph DI

Information is displayed using bars.

Practice:

- comparing values
- percentage increase/decrease
- ratio
- total

Tip

Check the scale carefully.

A bar graph may use:

1 unit = 10

instead of:

1 unit = 1

22.3 Line Graph DI

Best for understanding trends over time.

Common questions:

- increase/decrease
- difference
- average
- growth percentage

Trick

Look for the actual values before comparing the visual height of the lines.

22.4 Pie Chart DI

A full circle = 360°

Therefore:

$\text{Percentage} = \text{Angle} / 360 \times 100$

Or:

$\text{Value} = \text{Total} \times \text{Angle} / 360$

Useful conversions

$$90^\circ = 25\%$$

$$180^\circ = 50\%$$

$$120^\circ = 33.33\%$$

$$72^\circ = 20\%$$

$$36^\circ = 10\%$$

22.5 Caselet DI

No graph is provided.

Instead, information comes in a paragraph.

Example:

A company has 500 employees. 40% are women. 25% of women work in HR...

You need to extract the information yourself.

Strategy

1. Read the question first.
2. Extract only relevant numbers.
3. Make a mini-table.
4. Solve.

22.6 Mixed DI

May combine:

- table + bar
- pie + table
- line + table

Tip

This looks complicated, but usually uses the same arithmetic you already learned.

PART 23 — Data Sufficiency

What is it?

You are not necessarily asked:

■ "What is the answer?"

You are asked:

■ "Is the information enough to answer the question?"

Example:

What is the age of A?

Statement I gives one fact.

Statement II gives another fact.

You determine whether:

- I alone is sufficient
- II alone is sufficient
- both together are sufficient
- either alone is sufficient
- neither is sufficient

Trick

Do not solve more than necessary.

You only need to know whether the information is **sufficient**.

PART 24 — Quantity Comparison

You compare:

Quantity I vs **Quantity II**

Possible result:

- $I > II$
- $I < II$
- $I = II$
- Cannot determine

Tip

Look for a simpler transformation instead of calculating both values fully.

PART 25 — Probability

Basic Idea

Probability measures how likely something is to happen.

$$\text{Probability} = \frac{\text{Favourable outcomes}}{\text{Total possible outcomes}}$$

Example:

A fair coin:

Probability of Head:

$\frac{1}{2}$

Important Rules

Probability is always between:

0 and 1

or:

0% and 100%

Complement

$$P(\text{Not } A) = 1 - P(A)$$

This is one of the most useful shortcuts.

Example:

Probability of rain = 0.3

Probability of no rain:

$$1 - 0.3 = 0.7$$

PART 26 — Permutation and Combination

Basic Idea

Permutation

Arrangement matters.

Example:

ABC and ACB are different arrangements.

Combination

Selection matters, order does not.

Example:

Selecting A and B is the same as selecting B and A.

Formulas

$$nPr = n! / (n-r) !$$

$$nCr = n! / [r! (n-r) !]$$

Memory Trick

P = Position

C = Choose

Permutation → arrangement

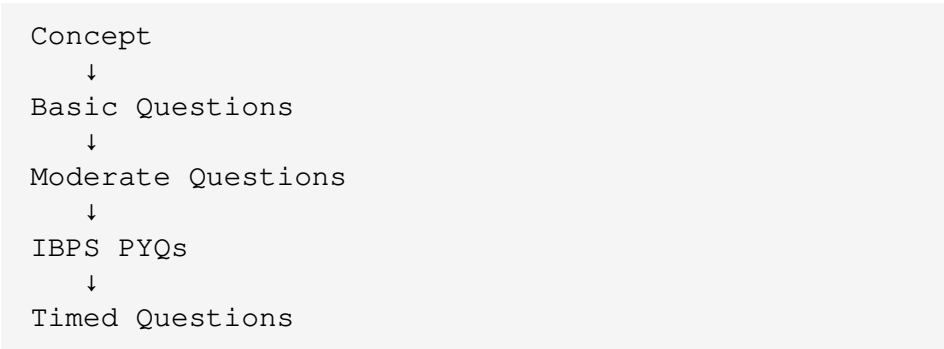
Combination → selection

This is lower priority than arithmetic for IBPS SO.

PART 27 — PYQ Preparation

After completing the concepts, begin previous-year questions.

For each topic:



Create a mistake log:

Topic	Mistake	Reason	Fix
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Percentage	Wrong base	Used new value	Always identify original value
Time & Work	Added times	Rates must be added	Convert to work rate
DI	Calculation error	Rushed	Slow down first
SI/CI	Wrong period	Forgot compounding	Check time + rate

PART 28 — Speed Tricks

Trick 1 — Fraction Thinking

Instead of calculating:

25% of 800

think:

$1/4$ of 800 = 200

Trick 2 — Convert Percentage to Fraction

For common percentages:

20% → 1/5
25% → 1/4
50% → 1/2

$$75\% \rightarrow 3/4$$

$$12.5\% \rightarrow 1/8$$

This is much faster.

Trick 3 — Use Base 100

When percentage changes are complicated, assume the original value is 100.

Example:

Increase by 20%, then decrease by 10%.

Start with:

$$100 \rightarrow 120 \rightarrow 108$$

$$\text{Net increase} = 8\%$$

Trick 4 — Use LCM in Time & Work

If A completes work in 12 days and B in 18 days:

$$\text{LCM of 12 and 18} = 36.$$

Think:

- A's daily work = 3 units
- B's daily work = 2 units
- Together = 5 units

$$\text{Total work} = 36 \text{ units.}$$

$$\text{Time} = 36/5 \text{ days.}$$

This is often easier than working directly with fractions.

Trick 5 — Use Options

In some questions, you do not need to fully solve the problem.

If options are widely separated, estimation can eliminate most options.

Trick 6 — Read Questions Before Data

In DI:

Question first → Data second.

Do not calculate everything just because the table contains it.

Trick 7 — Unit Check

Before solving:

- ₹ vs ₹000
- km vs m
- hours vs minutes
- percentage vs decimal

Many wrong answers are caused by unit mistakes, not difficult mathematics.

PART 29 — Common Exam Traps

Trap 1 — Wrong Percentage Base

"Increase by 20%" is based on the **original value**, unless the question says otherwise.

Trap 2 — Average Speed

Average speed is usually **not**:

$$(\text{Speed 1} + \text{Speed 2}) / 2$$

unless the conditions make that valid.

Trap 3 — Profit Percentage

Profit % uses **CP**, not SP.

Trap 4 — Discount Percentage

Discount % uses **MP**, not CP.

Trap 5 — Time & Work

Do not add the number of days.

Add the **work rates**.

Trap 6 — Upstream/Downstream

Downstream = Boat + Stream

Upstream = Boat - Stream

Trap 7 — DI Scale

Always check the axis scale.

PART 30 — How to Measure Progress

For each topic, use this system:

Level 1 — Concept

Can she explain the idea in simple English?

Level 2 — Basic

Can she solve straightforward questions?

Target: **90%**

Level 3 — Exam

Can she solve IBPS-level questions?

Target: **80–85%**

Level 4 — Speed

Can she solve accurately under time pressure?

Level 5 — PYQ

Can she recognise common patterns?

Recommended Topic Priority

Must Master

- Percentage
- Ratio & Proportion
- Average
- Profit & Loss
- Simple Interest
- Compound Interest
- Time & Work
- Time, Speed & Distance
- Simplification
- Approximation

- Number Series
- Data Interpretation

● Strong Preparation

- Mixture & Allegation
- Number System
- HCF & LCM
- Remainders
- Quadratic Equations
- Partnership
- Data Sufficiency
- Probability

● Lower Priority

- Ages
 - Permutation & Combination
 - Advanced remainder problems
 - Advanced probability
-

Recommended Study Sequence

Follow this exact order:

01. Number Basics
02. Fractions & Decimals
03. BODMAS / Simplification
04. Approximation
05. Percentage

06. Ratio & Proportion
07. Average
08. Profit, Loss & Discount
09. Simple Interest
10. Compound Interest
11. Partnership
12. Time & Work
13. Pipes & Cisterns
14. Time, Speed & Distance
15. Trains
16. Boats & Streams
17. Mixture & Allegation
18. Ages
19. Number System
20. HCF & LCM
21. Remainders
22. Quadratic Equations
23. Number Series
24. Table DI
25. Bar Graph DI
26. Line Graph DI
27. Pie Chart DI
28. Caselet DI
29. Mixed DI
30. Data Sufficiency
31. Quantity Comparison
32. Probability
33. Permutation & Combination
34. Topic-wise PYQs
35. Sectional Tests
36. Full Mocks

Final Rule

For IBPS SO HR, **Quant is a supporting subject, not the main subject.**

The ideal goal is:

Become comfortable enough with Quant to score reliably and quickly, while spending the biggest preparation effort on Professional Knowledge (HR, IR and Labour Laws).

A good preparation ratio is approximately:

50% HR Professional Knowledge + 20% Reasoning + 15% Quant + 10% English + 5% Mock/PYQ analysis initially, gradually increasing mock practice as the exam gets closer.